

Evaluating ML-KEM at (much) higher security levels

Swechchha Adhikari^{1*}, Vadim Lyubashevsky² and Peter Schwabe^{3,4}

¹ Brigham Young University, Provo, USA, adhikar6@byu.edu

² IBM Research, Zurich, Switzerland, VAD@zurich.ibm.com

³ Max Planck Institute for Security and Privacy, Bochum, Germany, peter@cryptojedi.org

⁴ Radboud University, Nijmegen, The Netherlands

“Conservative” post-quantum security could mean avoiding algebraic structure vulnerable to future attacks [BCD⁺16], or maintaining a wide security margin against improving cryptanalysis. FrodoKEM, standardized by ISO [ISO26], provides the first by design, but widening its margin requires scaling parameters that already cost nearly ten times the bandwidth of NIST-standardized ML-KEM at the same security level [Nat24, GLN⁺25]. We show a module scheme can buy this margin cheaply: KAIBURR reaches a large security margin within FrodoKEM’s bandwidth while running roughly 3× faster. Starting from ML-KEM-1024 and fixing every other parameter, we vary the rank k and measure the decryption-failure probability using the Kyber scripts [CRY]. The 2^{-128} bound holds at $k = 7$ but fails at $k = 8$, so raising k further cannot buy higher security without sacrificing correctness.

Instead of changing the ring dimension n or modulus q , we propose a parameterized noise distribution $f(t)$ on $\{-2, \dots, 2\}$ with $\Pr[\pm 2] = 2^{-t}$, $\Pr[\pm 1] = \frac{1}{4}$, $\Pr[0] = \frac{1}{2} - 2^{-(t-1)}$. Its standard deviation $\sigma(t) = \sqrt{0.5 + 2^{3-t}}$ decreases from 1 to $\sqrt{0.5}$ as t grows. A sample is drawn from uniform bits $a_1, \dots, a_t$ as $x = (2a_t - 1)(1 - a_1 + 2 \prod_{i=1}^{t-1} a_i)$, which is branchless; we machine-check both CT and speculative constant-time (SCT) in Jasmin [ABB⁺17].

Larger k raises decryption error, so we jointly raise t (lower σ) to compensate, while favoring byte-friendly values of t . $t = 4$ recovers ML-KEM’s sampler CBD(2), $t = 8$ allows one-byte-per-coefficient SIMD packing, and $t = 6$ gives more security per dimension and fewer PRF bytes at the cost of less regular packing. For each t we take the largest k meeting the failure bound and a communication cost under FrodoKEM-640 (9616 B pk, 9752 B ct).

Table 1: KAIBURR parameter sets. Classical/quantum security (Core-SVP) and failure probability computed with the Kyber scripts [CRY]; other parameters as in ML-KEM, compression removed.¹

Scheme	k	noise	failure	class.	quant.	pk (B)	ct (B)
kaiburr4	7	$f(4)$	$2^{-133.7}$	479	434	2,720	3,072
kaiburr6	18	$f(6)$	$2^{-134.1}$	1281	1162	6,944	7,296
kaiburr8	24	$f(8)$	$2^{-139.9}$	1717	1557	9,248	9,600

KAIBURR has reference and AVX2 implementations in both C and Jasmin, available [here](#). We benchmark on two Intel generations following the methodology of [ABB⁺23]: single-core with Hyper-Threading and Turbo Boost disabled, medians over 10 001 runs with randomness from Linux `getrandom` (Table 2).

Table 2: KAIBURR (AVX2) vs. FrodoKEM-640-SHAKE (AVX2), CPU cycles; KAIBURR8 runs ~ 2.6 – $3.1\times$ faster than Frodo. Jasmin stays within 17% of C, faster on Coffee Lake and slower on Raptor Lake.

Microarch.	Scheme	Impl.	keypair	enc	dec	
Coffee L. (i7-8700)	kaiburr4	C	155,113	159,177	171,924	
		Jasmin	144,489	145,158	150,135	
	kaiburr6	C	806,319	815,330	842,327	
		Jasmin	762,472	770,586	781,276	
	kaiburr8	C	1,368,794	1,378,658	1,416,170	
		Jasmin	1,294,067	1,303,213	1,311,650	
	FrodoKEM-640-SHAKE ²		[GLN+25]	3,898,666	4,063,179	3,939,567
	Raptor L. (i9-13900K)	kaiburr4	C	120,886	124,488	134,556
Jasmin			141,676	143,442	148,714	
kaiburr6		C	647,542	651,956	676,062	
		Jasmin	746,326	750,672	763,608	
kaiburr8		C	1,095,258	1,103,782	1,133,918	
		Jasmin	1,265,650	1,271,618	1,285,842	
FrodoKEM-640-SHAKE		[GLN+25]	3,121,248	3,246,262	3,189,934	

We also implement and report KAIBURR on Pavona’s Asymmetric Cryptography Coprocessor (ACC) [Glo26], which reuses ML-KEM’s arithmetic core. Our simulation model raises DMEM from 32 to 128 KiB for large k .

Table 3: Per-operation cost of KAIBURR on ACC, in clock cycles; single deterministic cycle-accurate simulator run.

Scheme	keypair	enc	dec
kaiburr4	240,816	239,348	257,474
kaiburr6	1,453,490	1,424,342	1,467,109
kaiburr8	2,505,793	2,449,864	2,506,070

KAIBURR shows that a structured KEM can reach conservative security margins cheaply and within FrodoKEM’s bandwidth, across software and dedicated PQC hardware.

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¹Removing compression allows us to obtain formally verified correctness bounds as in [BKHL⁺25]; this is left as future work.

²We modified the Frodo harness (<https://github.com/microsoft/PQCrypto-LWEKE>) to take the median of 10 001 runs for consistency.

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